

3. Explain the following concepts:

- a) Adaptive Expectations
- b) Rational Expectations
- c) Non-Accelerating Inflation Rate of Unemployment (NAIRU)
- d) Long-Run Phillips Curve

.....

.....

.....

.....

.....

.....

.....

6.6 LET US SUM UP

Unemployment results in loss of not only potential output at the macro level but also in income at the individual level. Many a time unemployment culminates into a crisis situation when there is widespread unemployment in the economy. The social stigma and psychological trauma associated with unemployment often compels policy makers to cut down on the rate of unemployment.

The classical economists assumed flexibility in real wage and prices which ensured full employment in the economy all the time. Keynesian economists, however, contest such an assumption and speak about rigidities in wage rate and prices. In case of sticky prices there is a possibility of unemployment as per the Keynesian model.

Phillips curve describes the inverse relationship between inflation and unemployment. It shows the possibility that unemployment can be reduced at the cost of higher inflation.

During the 1970s most economies in the world passed through a phase of stagflation. The trade-off between inflation and unemployment was proved to be false. In order to explain such a situation we bring in expectations into our analysis. There are two models of expectations: adaptive and rational.

According to adaptive expectations, the Phillips curve is stable in the short-run but in the long run it shifts. The long run Phillips curve is vertical. Thus there could be some trade-off between inflation and unemployment in the short-run, but in the long-run there is no trade-off. We explained the process through which the shift in the Phillips curve takes place. According to rational expectation, there is no trade-off between inflation and unemployment. Any policy of the government to reduce unemployment becomes ineffective, as people can forecast the expected changes correctly.

6.7 ANSWER/HINTS TO CHECK YOUR PROGRESS EXERCISES

**Inflation and
Unemployment**

Check Your Progress 1

1. Your answer should include frictional, structural and cyclical unemployment. Go through Section 6.2 for details.
2. Go through Section 6.3 and refer to Fig. 6.1.
3. These concepts are discussed in Sections 6.2 and 6.3.

Check Your Progress 2

1. Refer to the text in Section 6.5. You should explain Fig. 6.2.
2. These concepts are defined in Sections 6.5.



UNIT 7 RATIONAL EXPECTATIONS *

Structure

- 7.0 Objectives
- 7.1 Introduction
- 7.2 AS-AD Model and the Non-neutrality of Money
- 7.3 The Lucas Critique
- 7.4 Perfect Foresight and the Neutrality of Money
- 7.5 Rational Expectations Hypothesis (REH)
- 5.6 Lucas Supply Function
- 5.7 Policy Ineffectiveness Theorem
- 7.8 Let Us Sum Up
- 7.9 Answer/Hints to Check Your Progress Exercises

7.0 OBJECTIVES

After going through this Unit, you should be in a position to

- explain the concept of rational expectations and differentiate it from adaptive expectations;
- explain the relationship between the AS-AD model and the non-neutrality of money;
- discuss the Lucas critique;
- bring out the relationship between perfect foresight and neutrality of money;
- describe the rational expectations hypothesis;
- discuss the Lucas supply function; and
- explain the policy ineffectiveness theorem.

7.1 INTRODUCTION

Several important decisions by economic agents are based on expected values of economic variables. Let us take, for example, the case of Friedman's permanent income hypothesis. According to Friedman consumption expenditure depends not on current income, but on permanent income. Thus a reduction in income tax will

lead to an increase in consumption expenditure only if the reduction in tax rate is expected to be permanent; not temporary. Similarly, how much labour a worker will supply to the market is based on his/her expectations of future real wages. While nominal wages are negotiated before the work is undertaken, they are received afterwards. Thus, real wages ($= \text{nominal wages} / \text{aggregate price level}$) received by labour is known only in the future, when the worker goes to spend it at the prevailing price level of goods and services. An implication of the above is that how the future expected price level is determined is critical in decision-making. Expected price level determines expected real wages; it further determines whether a worker will continue to supply labour effort or withdraw it. Therefore, it is important to know the way expectations are formed.

Recall from Unit 1 that the classical economists assumed money to be neutral. Its implication is that a change in money supply affects nominal variables (e.g., price level) only while real variables (e.g., output, employment) remain unaffected. Most neo-classical economists, using the AS-AD approach, however realize that money does have an impact on real variables such as real wages, unemployment and output in the short run. This short-run impact of money arises due to the fact that there is a lag between expected value of real wages and actual value of real wages.

The initial impact of an increase in money supply is to increase the demand for goods and prices. Workers offer more labour, being 'fooled' into thinking that their real wages have increased, while actually real wages have not. In these 'fooling models', firms do not have any illusion – the firms realize that real wages have decreased, and therefore the firms hire more workers.

In this Unit we will discuss the non-neutrality of money in the short run using the AS-AD approach. Then, we will move on to the Lucas Critique which explains why agents are unlikely to form expectations that will persistently underestimate price rises. We will illustrate the results of the AS-AD model and contrast it with the case of 'perfect foresight', where neutrality of money would hold even in the short run.

We will discuss how expectations are formed using 'rational expectations'. As the name suggests, rational expectations refers to the value that a variable will take in the future. Rational expectations are formed on the assumption of rationality, i.e., based on incorporating all information about the variable and what its factors, i.e., knowing the true model explaining the variable. Then the variable is totally predictable except for random factors that cannot ever be anticipated. Uncertainty about the impact of random events or lack of global information can lead to less than 'perfect foresight', i.e., the variable not being totally predictable, or the expected value differing from the actual value in the future.

Rational expectations models come under the purview of new-classical economics. We know that in classical economics, there is perfect information, perfect flexibility in wages and prices and instantaneous market-clearing. These

features of classical economics ensure full employment in the labour market and corresponding full employment output. The new-classical economists allow for imperfect information (agents may have local information, not global information). Thus, workers and firms may not know the aggregate price level, although they would know their own wages and price levels. Unlike the AS-AD model, rational expectations models are not based on 'asymmetric information' across agents – workers and firms have access to the same information and are equally 'fooled' by unanticipated events. We will talk about this in details while discussing the Lucas Supply Curve. Finally, we will discuss about the 'policy ineffectiveness proposition'. This proposition uses the Lucas Supply Curve and the assumption of rational expectations to show that anticipated monetary policy is neutral with no effect on output and that only the unanticipated part of monetary policy impacts output. Thus it makes ineffective the use of any systematic monetary policy.

7.2 THE AS-AD MODEL AND NEUTRALITY OF MONEY

The aggregate demand (AD) function shows combinations of the aggregate price level (P) and real income (Y) which show equilibrium in the goods and money markets. It is a downward sloping curve reflecting that lower price levels are associated with higher real money balances and the resulting lower interest rates are associated with higher investment expenditure and therefore higher aggregate demand.

The AD function: $Y = f(G, T, M/P)$ is an increasing function of G, T and M/P, where G = Government expenditure, T = Taxes; M/P = nominal money supply/price level.

The aggregate supply (AS) equation shows labour market equilibrium – higher levels of output are associated with higher employment, higher wages due to the tighter labour market and, with cost plus pricing, imply higher prices.

The AS function comes from

- a) the wage function: $W = P^e f(u, z)$

where W = wage rate; P^e = expected price, u = unemployment rate, and z is a variable capturing all other factors affecting W

- b) the price setting relationship: $P = (1 + m)W$, where m = mark-up over cost (lower for a more competitive economy).

Hence, the AS function is $P = P^e (1 + m) f(u, z)$

Or, $P = P^e (1 + m) f(1 - Y/L, z)$

where u is replaced by $(1 - Y)/L$ (assuming a simple production function $Y = N$ = number of employed out of the total labour force = L).

When $P = P^e$, the real wage is what firms and workers have to jointly determine in the labour market (given u and z), and the resulting rate of unemployment is the ‘natural rate of unemployment’

Now $P = P^e$ coincide in the long run as any gap between price and expected price decreases with workers reducing the expectation error over time. So the natural rate of unemployment prevails in the long run.

However, in the short run, any divergence between actual P and expected P implies that workers’ expected real wage level is not met and the AS curve will shift accordingly, leading to output levels changing from the ‘natural level of output’.

In Fig. 7.1, we start with aggregate demand curve (AD_0) and aggregate supply curve (AS_0). Output is at the natural level of output Y_N and price P_0 coincides with the expected price P_0 .

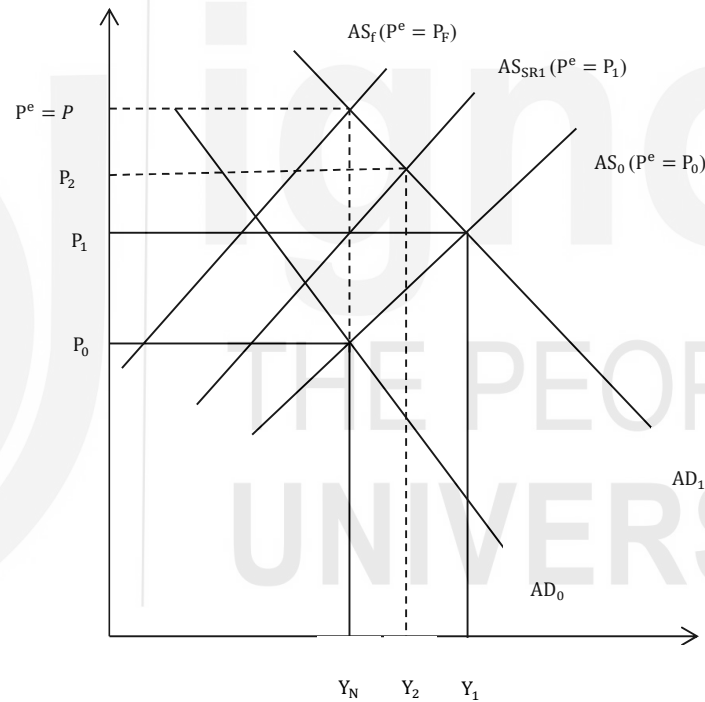


Fig. 7.1: The AS-AD Model without Rational Expectations

Let us assume that there is an increase in money supply. This will shift the AD curve to the right (from AD_0 to AD_1) and result in higher output Y_1 and a higher aggregate price level P_1 (intersection of the AS and AD curves). Thus, in the short run (where P^e is fixed), the increase in the money supply has caused output to change and we see that money is not neutral. This short run impact of money comes from the lack of anticipation by workers on the future impact of the increased money supply on prices and real wages, i.e., due to the miss-match of P and P^e . The initial impact of the increase in money supply is to increase demand for goods, demand by firms for more labour and nominal wages. However, workers do not anticipate that prices will also rise and that while nominal wages

have risen, real wages have fallen since P rises more than P^e (which initially is constant and on what the AS curve depends). Therefore, workers offer more labour effort, being ‘fooled’ into thinking that their real wages have increased.

The traditional Phillip’s Curve showing the trade off between price rise and unemployment was based on the same phenomenon of assuming that price expectations of workers are unchanged even when a price rise is observed.

Only when workers spend their wages and realize that the price level (P_1) is higher than they had expected (P_0), they realize that the real wages have actually declined. Then the new wage contracts will demand higher nominal wages, and the AS curve shifts up to AS_2 . Here again, actual prices are higher than the expected prices P_1 , output is lower at Y_2 but still higher than Y_N . The economy is still in the short run. The process continues iteratively, till the AS curve is finally at AS_f , where $P = P^e$ again, workers are getting their desired real wage, and output is back at Y_N .

This is the same as what the ‘modified’ Phillip’s Curve shows: price expectations rise when price is greater than expected price, so only a price rise higher than expected will lead to an increase in employment or output, Y . Notice that in these models, firms unlike workers realize that real wages have declined and therefore hire more workers. Thus, there is asymmetric information on the part of workers and firms.

In the above model, workers’ expectations of the price level are based on the past period’s price level. In other words, expectations of workers is $P_t = P_{t-1}$. This is a form of adaptive expectations where expectations are changed according to past observed actual values.

7.3 THE LUCAS CRITIQUE

The concept of rational expectations was introduced by John Muth in 1961. Robert E. Lucas and Thomas Sargent introduced this concept in economics through their writings in 1970s.

In the AS-AD model discussed earlier, money is neutral in the long run (when $P = P^e$). In the short run, however, workers think that their nominal wage increase is a real wage increase. Thus, in the short run, money can have real impact on employment and output.

Suppose, the policy maker increases the growth rate of money supply, as a result of which there is a price rise. The expected price is P_0 , but actual price turns out to be P_1 . Economic agents adapt their expected price to be P_1 in the next period. Actual price however increase further to P_2 . So, even though workers do not realize it (since their expectations of price are taking time to catch up to the actual price), real wages will fall – there is short run impact of money.

Lucas questions this ‘fooling’ of workers after past monetary policy has been used to increase employment. Having observed similar episodes of rising prices after increases in money supply in the past, should not economic agents raise

their expected price? They should take lessons from the past along with all relevant information in order to make rational predictions. A sensible forecast of price should be made on the basis of the behavior of the actual economy i.e. understanding the process or true model that generates the actual price and any new relevant information that comes up.

Thus, Lucas questioned the premise that people's expectations would remain unchanged when policy makers take advantage of a relationship based on past data on employment and inflation. The past relationship was based on people's average expectation of inflation remaining unchanged at zero. If policy makers have been increasing money supply to increase employment, resulting in a persistent price increase, people's expectations of inflation are now likely to increase, with the result that the increased money supply leads to higher prices but not higher employment.

This is the 'Lucas critique' – past statistical relationships in data are based on a certain expectations, when the policy environment changes, expectations are likely to change and the statistical relationships displayed in the past may no longer be hold.

As another example, let us take the case of policy makers decreasing the growth of money supply. The objective of the policy maker is to reduce inflation. Suppose, people base their expectation of future prices on last year's higher prices. Actual prices, however, would be lower due to decline in money supply growth rate. This will result in higher real wages and lower employment.

In the above example, suppose the government announces that it would decrease inflation by reducing the growth of money supply (and people believe the government because of its past records). Consequently, people with rational expectations will incorporate this relevant new information, and reduce their expectations of the price level. Accordingly, the nominal wages would decrease. Therefore, when actual prices fall, there is no increase in real wage and unemployment.

According to Lucas, people make rational forecasts on variables (such as prices) on the basis of policy environment and available information. Lucas' critique is that when there is a change in the policy environment, economic agents will revise their expectations.

7.4 PERFECT FORESIGHT AND THE NEUTRALITY OF MONEY

There is the 'perfect foresight' if we know the AS-AD model equations and parameters that determine the actual price, P . If we use this model to get the value of P^e (expected price), the P^e should equal P . That means there will be no expectational error ($P - P^e = 0$). Consequently, workers would not be fooled into thinking that an increase in nominal wages is an increase in real wages.

In that case, there would be no increase in employment that would result from the increase in money supply and money would be neutral. Note that the above also requires that wages and prices are totally flexible, so that the real wage can be maintained at the desired level at all times and employment remains at the natural rate of unemployment. Thus, it is very similar to the Classical Model with regards to its performance.

In Fig. 7.1, if $P^e = P$, then the level of output would remain at Y_N . In the case of perfect foresight, price would increase proportionate to the increase in money supply immediately, real wages would remain unchanged and there is no change in employment level. There would be no role for monetary policy to play in this case because every policy that would be implemented would be immediately understood by the public. As a result, only nominal variables would be affected and real variables will remain unchanged.

However, in the real world, especially for economic variables which are impacted by human behaviour with random elements, there is no 'perfect foresight'. At best, people can make 'rational expectations' on which to base decisions.

Check Your Progress 1

1. Describe the AS-AD model. Illustrate the condition of natural rate of unemployment in a diagram.

.....

.....

.....

.....

2. Illustrate the impact of an increase in money supply in the short-run through the AS-AD diagram.

.....

.....

.....

.....

3. Explain the Lucas critique on the effectiveness of monetary policy.

.....

.....

.....

.....

4. What is meant by perfect foresight? Under such condition, how do the changes in money supply affect an economy?

.....

.....

.....

.....

7.5 RATIONAL EXPECTATIONS HYPOTHESIS

In this section we will discuss as to how rational expectations are formed. Also we will discuss the properties and implications of rational expectations hypothesis (REH). We will start with a model that depicts ‘perfect foresight’ and then contrast it with a rational expectations model.

The future value of a variable, e.g., prices, interest rates or output, can be predicted or guessed in several ways – through astrology, throwing darts, based on past trends, adapting (changing) or correcting past expectations using more recent data on the variable. However, the above approaches are unlikely to generate good predictions even on average because they ignore the underlying process that influences the variable.

The method of rational expectations assumes that individuals have observed enough history on the concerned variable to understand its explanatory factors, and the systematic manner in which these explanatory factors influence the variable. Thus, individuals use their knowledge about the process of generating the variable. These systematic patterns then restrict the values that the variable can take. This helps individuals in forming their expectations about the likely future value (or expected value) of the concerned variable.

As an example, using a general algebraic representation, we assume that the ‘true model’ (process that generates variable Y) is known with the relevant variables affecting it are X , Y and Z . The relationship between the variables given as follows:

$$Y_t = \alpha + \beta X_{t-1} + \gamma Y_{t-1} + \eta Z_{t-1} \quad \dots (7.1)$$

where Y_t is the actual value of Y at time t and is a function of observed values at time $(t-1)$ of variables X , Y and Z . Assume that α is a constant term, and β , γ and η are coefficients of observed values of X , Y and Z .

Equation (7.1) is a deterministic model where the right hand side variables completely determine the value of Y . Now, the mathematical expectation of Y_t is:

$$E_{t-1} Y_t = \alpha + \beta X_{t-1} + \gamma Y_{t-1} + \eta Z_{t-1} \quad \dots (7.2)$$

where $E_{t-1} Y_t$ is the expected or predicted value of Y made at the end of time period $(t-1)$ for the future time period t based on observed values at time $(t-1)$ of variables X , Y and Z .

Equation (7.1) is also a model where agents have ‘perfect foresight’ and can accurately predict the value of Y . Thus, expectational error = [actual value of Y (that is, Y_t) – expected value of Y (that is, Y_t^e)] = 0 in the case of equation (7.1). As an example, with perfect foresight, the expected price level following an increase in the money supply would be an equi-proportionate increase in the price level.

Let us try to re-formulate equation (7.1) in terms of rational expectations. The actual value of Y in period t depends not only on the variables X , Y and Z and